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Forecasting unemployment insurance claims in realtime with Google Trends[☆]

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ABSTRACT

Leveraging the increasing availability of "big data" to inform forecasts of labor market activity is an active, yet challenging, area of research. Often, the primary difficulty is finding credible ways with which to consistently identify key elasticities necessary for prediction. To illustrate, we utilize a state-level event-study focused on the costliest hurricanes to hit the U.S. mainland since 2004 in order to estimate the elasticity of initial unemployment insurance (UI) claims with respect to search intensity, as measured by Google Trends. We show that our hurricane-driven Google Trends elasticity leads to superior real-time forecasts of initial UI claims relative to other commonly used models. Our approach is also amenable to forecasting both at the state and national levels, and is shown to be well-calibrated in its assessment of the level of uncertainty for its out-of-sample predictions during the Covid-19 pandemic.

1. Introduction

Unemployment insurance (UI) claims are the single most timely indicator of U.S. labor market activity, making them a widely tracked leading indicator of the business cycle.[1] Furthermore, because UI claims can be tracked at the state level, they generate considerable interest amongst local policymakers. This attention was particularly acute during the Covid-19 pandemic when the trajectory of the virus and its impact was at times geographically uneven (e.g., Goldsmith-Pinkham and Sojourner (2020), Larson and Sinclair (2021)).

The advent of “big data” has the potential to substantially improve the forecasting of official statistics like UI claims. One well-publicized example is Google Trends, which tracks word or phrase patterns entered into the Google search engine. Previous attempts to incorporate Google Trends into forecasts of UI claims have been of mixed success.[2] For instance, both the models of Choi and

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1 For example, UI claims are included in the Conference Board's Leading Economic Index as well as large-scale macroeconomic forecasting models (e.g., Brave et al. (2019)), and is closely followed by financial markets to gauge changes in payroll employment and the unemployment rate. See also Borup et al. (2020), Gordon (2009), Lewis et al. (2020).

2 In addition to UI claims, investigations are focusing on the unemployment rate by D'Amuri and Marcucci (2017) for the U.S.; McLaren and Shanbhogue (2011) for the U.K.; Askitas and Zimmermann (2009) for Germany; Suhoy (2009) for Israel; Tuhkuri (2016) for Finland; and, Capema et al. (2020) for the EU.

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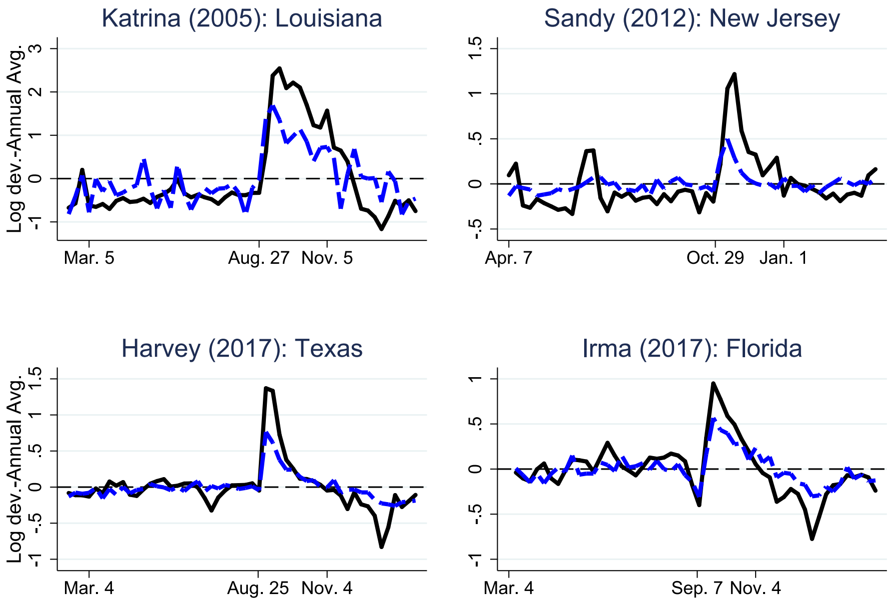

Fig. 1. Initial claims and Google unemployment searches around four hurricanes. *Notes*: This figure displays the time series of the log share of initial unemployment insurance claims (black line) and the log ratio of the Google Trends measure relative to the U.S. (blue dashed line) for the "unemployment" topic less the annual average for the six months leading up to and following the landfall for the primarily affected state of each of our four most damaging hurricanes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Varian (2012) and Larson and Sinclair (2021) were not able to best a univariate AR(1) model.

Many previous attempts to forecast UI claims with Google Trends rely purely on national search patterns to identify the key take-up parameter for UI. For several reasons, this aggregation could be misleading. For example, the models typically incorporate a lagged dependent variable; in the context of UI claims, a notoriously persistent series, forecasting performance is likely being impaired by finite sample bias exacerbated by the limited history of Google Trends. We present evidence in support of this fact for the seminal Choi and Varian (2012) model. This bias is potentially further compounded when a Google search of an "unemployment" related term might not only indicate an increased likelihood of someone claiming unemployment insurance benefits, but also interest in the news about future labor market conditions.[3]

Moreover, the existing literature that makes use of state-level data (e.g., Larson and Sinclair (2021)) has largely avoided an additional series of identification issues. Using state-level UI data raises an altogether different set of issues than a time series regression with national UI data. For instance, the weekly state-level UI data are not seasonally adjusted, exhibit stochastic trends, are influenced by the individual state rules governing UI programs (and their changes over time), and are likely to be more sensitive to measurement error and revisions depending on the size of the state. We propose an empirical strategy that can overcome these hurdles.

Our key insight is that natural disasters like hurricanes generate sharp spatial changes in new UI claims that are picked up in Google search activity (e.g., see Fig. 1 for four of our selected hurricane events) and, critically, are unrelated to news about future national labor market conditions. This sharp local variation is also helpful for reasons that pertain to using state-level UI claims data. First, it allows us to "difference-out" variation associated with seasonality and stochastic trends by way of a suitable control group. Second, it permits us to accommodate state-level heterogeneity in UI programs by identifying the elasticity primarily off within-state variation. Third, it allows us to use relatively short event windows which modulates concerns regarding idiosyncratic changes in individual state UI systems.

We apply this insight to all of the major hurricanes to make landfall in the mainland U.S. since 2004 to show that Google Trends is highly predictive of subsequent initial UI claims at the state level. Using parameters derived from these hurricane periods, we demonstrate that the evolution of initial UI claims during the 2008–09 and 2020 recessions was largely predicted by Google Trends. Furthermore, we show that our hurricane-based elasticity outperformed common univariate forecasting models for UI claims as well as the model of Choi and Varian (2012).

In the case of the most recent Pandemic Recession, we further demonstrate that our estimated elasticity is useful for predicting new claims at both the national and state level, and is well-calibrated in its assessment of the level of uncertainty for its out-of-sample predictions. Additionally, through a handful of state-level case studies, we show how our approach also highlights the presence of other potential measurement concerns that are most salient to the state-level UI claims data, including

[3] To the extent that there is also measurement error in the Google Trends series, the estimated relationship might additionally suffer from attenuation bias. One possible source of this measurement error could come from the fact that Google Trends itself represents a sample. For more discussion surrounding this issue, see Section 2.

expansions to UI eligibility (e.g., through the Pandemic Unemployment Assistance (PUA) program), processing delays, and fraud–all of which plagued policymakers relying on state and national UI claims data during the pandemic (New York Times, 2020).

Our results contribute to three pieces of literature. First, we add to research that leverages Google Trends data to inform forecasting models of economic indicators. Similar to Choi and Varian (2012), the focus of our paper rests solely on predicting initial UI claims. However, Google Trends has been used to study a variety of economic indicators, including GDP (Ferrara & Simoni, 2019; Woloszko, 2020), home prices and building permits (Beracha & Wintoki, 2013; Coble & Pincheira, 2017), consumption (Vosen & Schmidt, 2011), automobile purchases in emerging markets (Carriere-Swallow & Labbe, 2013), and the global demand for oil (Yu et al., 2019). Second, we contribute to a growing literature that uses hurricanes to identify key structural elasticities that can inform policy decisions (e.g., see Baker et al. (2020), Deryugina (2017), Deryugina et al. (2018), Gallagher and Hartley (2017), Ortega and Taşpinar (2018)). Finally, our results contribute to a now giant body of work from a diverse set of disciplines investigating the economic consequences of the Covid-19 pandemic, and, in particular, those tracking realtime developments in the labor market (e.g., see Cajner et al. (2020a, 2020b), Farrell et al. (2020), Ganong et al. (2020)).

2. Data

We use three sources of data: (a) the U.S. Department of Labor's weekly release of initial UI claims, (b) the weekly Google Trends index of unemployment-related search histories, and (c) a list of the costliest hurricanes to hit the US mainland since 2004.

2.1. UI claims

Initial UI claims are reported at the weekly frequency by the Department of Labor for reference weeks that start on Sunday and run through the following Saturday.[4] The first, or advance, release of the data is published with a five-day lag on the Thursday following the end of the reference week. The final release is published the following Thursday, with a 12-day lag from the end of the reference week. We use the final release of both national and state-level counts in addition to D.C.; the national time series from January 2004 to May 2020 is displayed in Fig. 2. While the Department of Labor reports a seasonally adjusted version of national initial UI claims, it does not seasonally adjust state-level measures.[5] Consequently, all of our analysis is conducted on non-seasonally adjusted data.

[4] The claims data can be found at U.S. Employment and Training Administration (2020).

[5] Additionally, on September 3, 2020, the Department of Labor revised its seasonal adjustment procedure on the national UI claims series moving from a multiplicative adjustment procedure to an additive one.

2.2. Google trends

Google Trends[6] indexes measure search activity for a particular term (like "jobs"), or a collection of related terms referred to as a *topic*, over any time period beyond four hours. We focus on search activity under the topic "unemployment" at a weekly frequency that matches the UI claims reporting periods. The strong correlation between UI claims and Google searches on unemployment, depicted in Fig. 2, makes clear the timeliness advantage of using Google Trends data at even the weekly frequency, given the lag that exists in UI reporting, particularly for policymakers interested in forecasting the realtime demand for UI.

There are both advantages and disadvantages to using a Google search topic rather than a search word. A broad topic avoids idiosyncrasies associated with the specific wording of queries. Take the search word "jobs" as one intuitive alternative. We can think of several problems of varying seriousness and generality of such a choice. First, jobs could be a term used for employed individuals searching on-the-job and therefore unlikely to file for unemployment insurance. Second, a specific term like jobs may hide the diversity of search activity that precedes a UI claim. Finally, singular search terms are prone to adding measurement error as a consequence of low propensity, but extreme, idiosyncratic events that are otherwise totally unrelated to the activity to be captured. Again using the same example, the death of Apple's co-founder Steve Jobs spurred an unprecedented spike in the search term "jobs".[7]

A clear drawback to using a broad Google Trends topic is the relative "black box" nature of its construction. At least in practice, the index for a specific term like "jobs" has a well-defined numerator (e.g., number of searches which included "jobs" as a part of the keywords) and denominator (e.g., all searches during that time period in that region). For the topic indexes, this level of transparency is lost. Moreover, perhaps the very nature of its construction has changed over time, as Google adjusts its search topic algorithms. While we are confident that the unemployment topic index makes for an appropriate input for our forecasting exercise with initial UI claims, not all of the topic indexes are likely superior to a more restrictive set of search terms.[8]

A further but minor consideration is when the weekly Google Trends unemployment topic index is made publicly available (e.g., see Larson and Sinclair (2021)). We have found that there is a small delay of no more than two days. Fortunately, introducing splicing techniques to

[6] See https://trends.google.com/trends/?geo=US.

[7] This may be a somewhat frivolous example but it is also worth pointing out that the death of Steve Jobs on October 5, 2011, would have fallen within even our most conservative event window for hurricane Irene. More generally, it hopefully highlights a potential risk of specific words.

[8] An alternative empirical approach to accommodating the possibly high dimensional set of Google Trends indexes that arise from an inclusive set of search terms is the use of machine learning and other dimension-reducing regularization methods (e.g., Borup and Schütte (2020)).

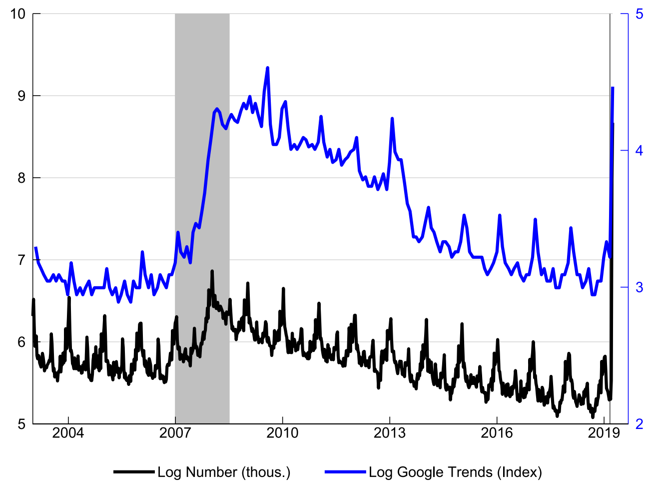

Fig. 2. Initial claims and Google unemployment searches, 2004-2020. *Notes*: This figure displays the time series of initial unemployment insurance claims (black, non-seasonally adjusted (nsa)) and the Google Trends index (blue, nsa) for the "unemployment" topic over the time period of January 2004 to May 2020. US recessions defined by the National Bureau of Economic Research are shaded in gray. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

deal with this delay does not significantly alter our results. But a conservative interpretation of our research design is that it produces a "backcast" of the UI claims number for a Sunday-Saturday week the *Monday* following that Saturday. Even this timing represents a meaningful advantage, especially around large turning points in the economy.

2.3. Hurricanes

Our research design relies on UI activity around the 11 costliest hurricanes (each causing at least $10 billion (in $2020) in damage) to hit mainland U.S. since 2004, when Google Trends became available: Katrina in 2005, Harvey in 2017, Sandy in 2012, Irma in 2017, Ike in 2008, Wilma in 2005, Michael in 2018, Rita in 2005, Florence in 2018, Irene in 2011, and Matthew in 2016.[9] The total damage, as estimated by the National Oceanic and Atmospheric Association (NOAA), and the approximate mid-point day of each hurricane is reported in columns (2) and (3) of Table 1, respectively.

For each hurricane, we select the state which NOAA estimates experienced the most damage as our "treated" state.[10] These states, reported in column (4), are Louisiana for Katrina; Texas for Harvey, Ike, and Rita; New Jersey for Sandy; Florida for Irma, Wilma, and Michael; and North Carolina for Florence, Irene, and Matthew. We also need a control state to account for pre-existing trends in labor market conditions. In our baseline, we use the entire U.S. as a control. Alternatively, to ensure that differences in state-level unemployment insurance generosity are not impacting our estimates, we identify the most comparable state in terms of the maximum allowable unemployment insurance benefit (as reported by Hsu et al. (2018)) that is otherwise located geographically as far as possible from the treated state, so as not to confound spillovers from the hurricane event. These control states are listed in column (5).[11] Finally, we also consider a "placebo" test for each event. Specifically, we "re-route" the hurricane to another state among the collection of our treated states that is likely to have experienced no direct effects of the event. Given the geographical distribution of our treated states, this amounts to sending hurricanes that hit the Southern states (e.g., Florida and Texas) to New Jersey, and hurricanes that hit the mid-Atlantic states (e.g., North Carolina and New Jersey) to Texas (column 6).

The data is organized into a series of 11 one-year event windows (columns 7 and 8) around hurricanes. The Google Trends data are normalized into an index such that the point in time with the highest search activity within a state over the one-year window has a value of 100, with other values scaled proportionally. For example, for Hurricane Katrina, we perform four separate data pulls: one for the treated state (LA), one for the control state (AZ), one for the placebo state (NJ), and one for the U.S. as a whole. We then use search interest for the entire country to normalize search interest across the three states, making the relative numbers directly comparable. Because the Google Trends indices are based on samples, we pull search interest for each event-state combination ten times and average the results to minimize the impact of sampling variability.[12]

To motivate the discussion of our identification strategy in the next section, we plot the total damage of each

9 See NOAA (2020). The damage caused by Hurricane Maria, which did not make it to the mainland U.S. but affected Puerto Rico and its neighboring islands, would make it the third costliest hurricane.

10 We assign treated areas to be states because UI data is only available consistently at that geography.

11 Online Appendix Figure A.14 shows the standardized maximum allowable unemployment insurance benefit of the treatment and control states.

12 Overall, the sampling variation is small compared to the size of the increases in search activity associated with the hurricane events. For each event-state cell, we calculated the coefficient of variation, range, and standard deviation of the values we pulled. The mean coefficient of variation across all of the cells was 0.026; the mean range was 3.4, and the mean standard deviation was 1.5.

Table 1
Summary of hurricane events.

Event	Damage ($2020)	Event date	Affected state	Control state	Placebo state	Start date	End date
Katrina	170B	8/27/2005	Louisiana	Arizona	New Jersey	2/2005	2/2006
Harvey	131B	8/27/2017	Texas	Michigan	New Jersey	2/2017	2/2018
Sandy	74B	10/30/2012	New Jersey	Washington	Texas	4/2012	4/2013
Irma	53B	9/9/2017	Florida	Missouri	New Jersey	3/2017	3/2018
Ike	37B	9/13/2008	Texas	Michigan	New Jersey	3/2008	3/2009
Wilma	26B	10/25/2005	Florida	Missouri	New Jersey	4/2005	4/2006
Michael	26B	10/10/2018	Florida	Missouri	New Jersey	4/2018	4/2019
Rita	25B	9/25/2005	Texas	Michigan	New Jersey	3/2005	3/2006
Florence	25B	9/14/2018	North Carolina	Oregon	Texas	3/2018	3/2019
Irene	16B	8/27/2011	North Carolina	Oregon	Texas	2/2011	2/2012
Matthew	11B	10/10/2016	North Carolina	Oregon	Texas	4/2016	4/2017

Notes: This table summarizes the eleven hurricanes used in our event study analysis, including the approximate mid-point of each hurricane's duration, the affected state used for each event, the control state with the most comparable UI system, the placebo state, and the +/−6-month window that comprises the event window. Of the costliest hurricanes to have occurred since 2004, Maria was excluded because it did not hit the mainland U.S.

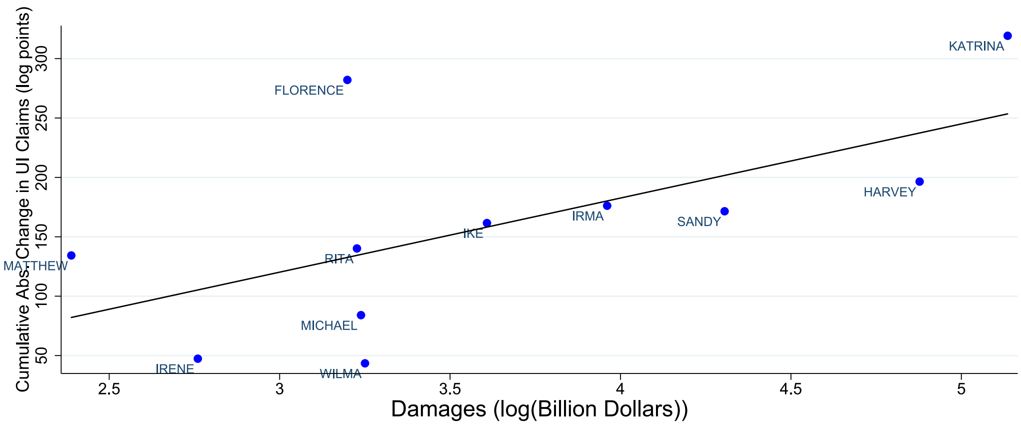

Fig. 3. Cumulative absolute change in UI claims and total hurricane damage. *Notes*: This figure displays the cumulative absolute log change of initial unemployment insurance claims for the treated state over the four week period following landfall versus the log total damages measured in 2020 dollars for each of the eleven hurricane-affected states: Louisiana (for Katrina), Texas (for Harvey, Ike, and Rita), New Jersey (for Sandy), Florida (for Irma and Wilma), and North Carolina (for Florence, Irene, and Matthew). We also plot the best linear fit line, which has a slope coefficient of (100×) 0.62 (standard error (100×) 0.24).

hurricane in the treated states against that state's cumulative four-week absolute log change of initial UI claims (in Fig. 3) and the Google Trends unemployment topic index (in Fig. 4). In both cases, the strength of the hurricane is highly positively associated with subsequent variation in UI claims and Google search activity. Katrina, Harvey, Sandy, and Irma represent the four most damaging hurricanes, and each of these led to dramatic week-to-week changes in new UI claims, which coincided with comparable variation in Google unemployment searches. While still causing in excess of $10 billion in damage, hurricanes Irene and Wilma led to more modest week-to-week changes in UI claims, with a similarly small effect on the variation in Google search activity. Our event study research design leverages this variation to credibly estimate the elasticity of UI take-up with respect to search intensity, which we will then in turn use to forecast the demand for unemployment insurance in real time during recessions.

3. Event studies

We measure the association between the Google Trends unemployment topic index and initial UI claims with the following linear regression equation:

$$\ln\left[\frac{\text{UIClaims}_{1ht}}{\text{UIClaims}_{0ht}}\right] = \alpha_h + \beta \ln\left[\frac{\text{GoogleTrends}_{1ht}}{\text{GoogleTrends}_{0ht}}\right] + \varepsilon_{ht}, \tag{1}$$

where h indexes a hurricane event and t indexes weeks within the year-long window around each hurricane. We accommodate seasonal and secular trends common across states by estimating the regression specification in terms of log-ratios, where the subscript 1 references the treated state and the subscript 0 denotes our control group. In our baseline specification, this control group is the U.S. To accommodate differences in the average number of UI claims for states of various population sizes, Eq. (1) also includes a set of event fixed effects (α_h).

To visualize the data and our parameter of interest, β, Fig. 5 presents a scatter plot of weekly UI claims shares against weekly Google Trends ratios net of event fixed effects. There is a positive and essentially linear association between the two when pooling all of the hurricane events together (as described by the best fit line) and within each

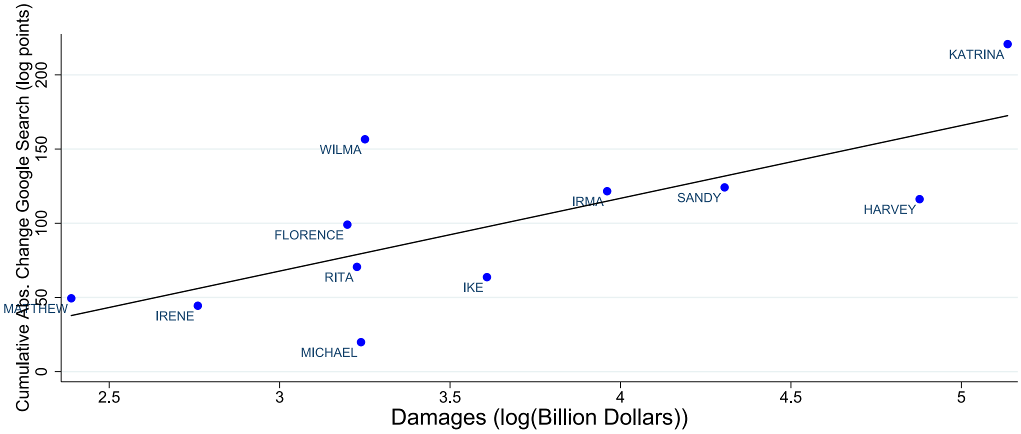

Fig. 4. Cumulative absolute change in Google Trends unemployment search and total hurricane damage. *Notes*: This figure displays the cumulative absolute log change in Google Trends search of the "unemployment" topic for the treated state over the four week period following landfall versus the log total damages measured in 2020 dollars for each of the eleven hurricane-affected states both net of a the annual average for the event: Louisiana (for Katrina), Texas (for Harvey, Ike, and Rita), New Jersey (for Sandy), Florida (for Irma and Wilma), and North Carolina (for Florence, Irene, and Matthew). We also plot the best linear fit line, which has a slope coefficient of (100×) 0.49 (standard error (100×) 0.15).

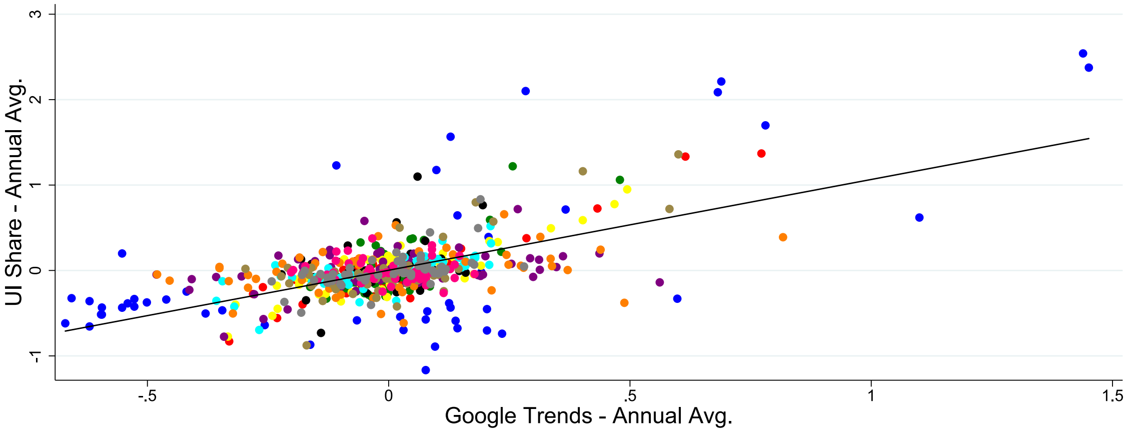

Fig. 5. UI claim shares and relative Google Trends. *Notes*: This figure displays the log share of initial unemployment insurance claims versus the log ratio of Google Trends for the "unemployment" topic for the eleven hurricane-affected states relative to the US: Katrina (Louisiana, in blue), Harvey (Texas, in red), Sandy (New Jersey, in green), Irma (Florida, in yellow), Ike (Texas, in black), Wilma (Florida, in purple), Rita (Texas, in orange), Florence (North Carolina, in brown), Irene (North Carolina, in pink), and Matthew (North Carolina, in gray). The solid line is the linear best fit line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

individual event too.[13] Indeed, that is what we find when we estimate Eq. (1).

Table 2 reports our baseline estimate of β is 1.06 with a standard error (estimated from 10,000 event-level bootstrap and jackknife samples) of 0.17. Given the precision of this estimate, we undertook an additional leave-one-out cross-validation of the regression to ensure that it was not overfitted to a particular hurricane. We report the estimates of the search elasticity from each of these additional specifications in columns (2) to (12). The point estimates that resulted from this exercise are in the range of [0.91, 1.17], within one standard error of the point estimate from our baseline specification.

The leave-one-out estimates are also useful for providing an out-of-sample metric of forecast accuracy. Averaging across all of the events, the *out-of-sample* R^2 is 0.53.[14] Fig. 6 presents the full set of out-of-sample one-week ahead forecasts over our hurricane sample, as well as their realized values. For ease of comparison, both the forecasted and realized values subtract annual averages of the forecasts at the state-event level and are scattered against a 45-degree line. Despite a few instances where errors are notable, the out-of-sample forecasts indicate that our regression model is well calibrated, a result which at least partly reflects the relative stability of the elasticity estimates in Table 2.

13 Including a quadratic term of the Google Trends (log-ratio) in our baseline specification leads to a coefficient on the quadratic term which we could not reject the null hypothesis was zero at the 0.01 significance level.

14 The root-mean-squared error (RMSE) of the out-of-sample forecasts generated by the leave-one-out cross-validation exercise is 0.29.

Table 2
Event study regression results.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Google Ratio	1.062	0.911	1.018	1.041	1.039	1.066	1.138	1.068	1.170	1.034	1.082	1.064
	0.177	0.229	0.189	0.183	0.193	0.182	0.174	0.189	0.148	0.192	0.178	0.183
Drop		Katrina	Harvey	Sandy	Irma	Ike	Wilma	Michael	Rita	Florence	Irene	Matthew
Adj. R^2	0.363	0.315	0.335	0.363	0.345	0.376	0.390	0.360	0.404	0.354	0.375	0.364
N	572	523	520	520	520	519	519	520	520	520	520	519

Notes: This table reports regression results for Eq. (1), where we report the coefficient on the Google Trends Ratio term and its standard error. Standard errors are estimated by 10,000 event-level bootstrap and jackknife samples. In column 1, we report estimates using all of the hurricane events, and each subsequent column reports results from leaving one of the eleven hurricane events out individually.

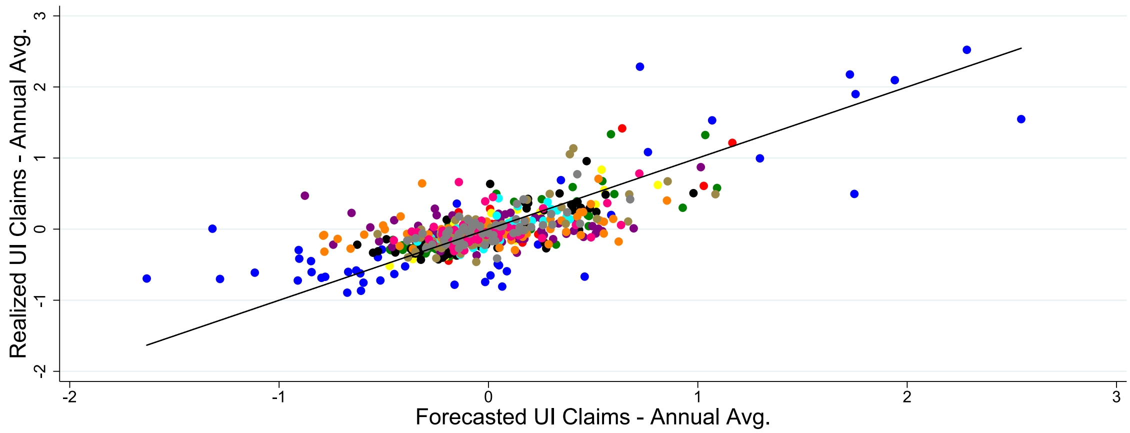

Fig. 6. Out-of-sample forecasts and realizations: Hurricane sample. *Notes*: This figure displays the scatter plot of the nowcast of initial unemployment insurance claims and the realized number of initial unemployment insurance claims for all weeks during each of our hurricanes using the estimated coefficient where the event being forecasted is left out of the sample. For each event Katrina (Louisiana, in blue), Harvey (Texas, in red), Sandy (New Jersey, in green), Irma (Florida, in yellow), Ike (Texas, in black), Wilma (Florida, in purple), Rita (Texas, in orange), Florence (North Carolina, in brown), Irene (North Carolina, in pink), and Matthew (North Carolina, in gray), the annual average of the forecast was taken out of both the realized values and the forecasted values. The line displayed on the graph is the 45-degree line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.1. Treatment effect heterogeneity

Our hurricane event study design was meant to alleviate the concerns that state-level differences in UI systems could confound our elasticity estimate, largely by using within event variation over relatively short time series lengths. Next, we explore the possibility that the elasticity of unemployment insurance take-up with respect to Google search activity varies significantly across our set of events. We explore this possibility along two dimensions: (i) the level of damage caused by the hurricane, and (ii) the treated state's generosity of unemployment insurance. This is a potentially important external validity issue; to the extent that the impact of the hurricane on the rate of UI take-up is highly heterogeneous – either for reasons related to the hurricane itself or the state that was impacted – the likelihood that our elasticity estimate will perform well in other settings is low. To investigate, we re-estimate Eq. (1) separately for each hurricane event – leaving us with eleven separate elasticity estimates.

Figs. 7 and 8 present the 11 independent elasticity estimates, one for each hurricane, against the amount of total damage inflicted by the hurricane and the generosity of the treatment state's UI program. There are several key takeaways from these figures. First, a modest amount of heterogeneity exists across our set of hurricanes, with the top and bottom decile ranging from [0.30, 1.68]. Second, while the estimated elasticities have a weakly positive association with the overall level of damage and state UI generosity, these relationships are not statistically significant. Thus, on balance, our baseline, pooled elasticity appears to be appropriate to use in other contexts. In the next section, we demonstrate that this conjecture is valid in the two most important settings of the Google Trends period: the 2008–09 and 2020 recessions (or, the Great Recession and Pandemic Recession as we refer to them in the next section).

3.2. Other robustness checks

To provide a more rigorous assessment of the stability of the elasticity estimates, we present a series of robustness checks in the Online Appendix. These include a more restrictive event window of +/- 3 months around hurricane landfall (Table A.8), allowing for a lag in the Google Trends ratio (Table A.5), and using a more conservative identification strategy – in particular, a version of the *identification through heteroskedasticity* approach (Table A.6 see Rigobon (2003)). We also consider an alternative control group based on the similarity of state UI program

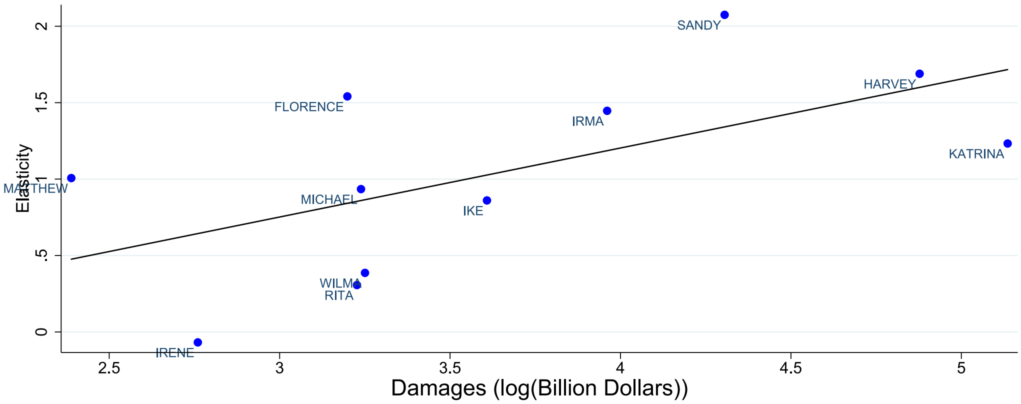

Fig. 7. Hurricane-specific UI elasticity versus total damage. *Notes*: This figure displays the hurricane-specific elasticity of Google trends search intensity on initial UI claims versus the log total damages measured in 2020 dollars for each of the eleven events. The solid line is the best linear fit, which has a slope coefficient of 0.45 (standard error 0.20).

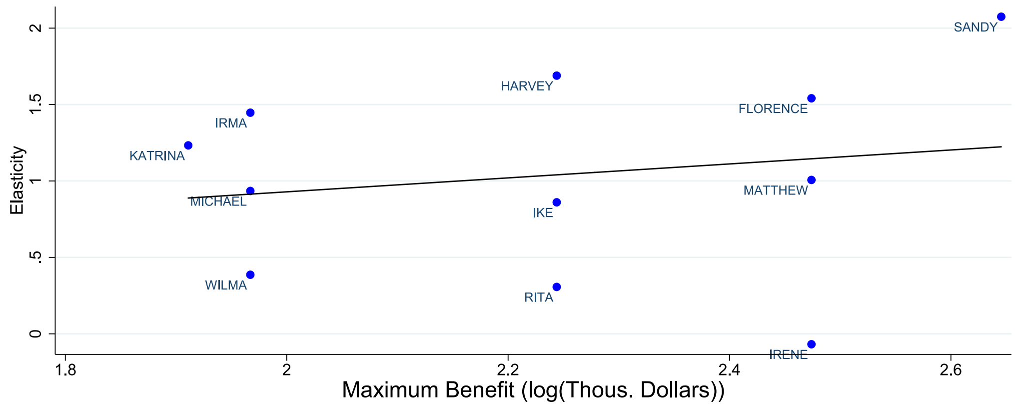

Fig. 8. Hurricane-specific UI elasticity versus state UI generosity. *Notes*: This figure displays the hurricane-specific elasticity of Google trends search intensity on initial UI claims versus the log of the state-specific maximum unemployment insurance benefit amount for each of the eleven events. The solid line is the best linear fit, which has a slope coefficient of 0.45 (standard error 0.86).

generosity (Table A.7).[15] Finally, we run a "placebo" test that assigns each hurricane event to one of the other treatment states in our study that is outside the affected region for that hurricane (Table A.9).[16] Finally, to demonstrate that our approach is also amenable to applications at finer geographical levels, we provide an elasticity estimate using metro-area UI claims and Google Trends data during the pandemic (Table A.10). Across all of these robustness checks, we find largely supportive evidence of our baseline elasticity estimate. To the extent that deviations exist, they are modest, often not statistically significant, and collectively in offsetting directions.

15 The chosen comparison states are listed in Table 1. The matching process is detailed in the online appendix. Briefly, we rescaled the maximum UI benefit measure of Hsu et al. (2018) to be in standard deviation units from the cross-sectional average for our sample period to facilitate making comparisons across states on the dimension of UI generosity. We then matched our treated states to a single control state based on UI generosity while ensuring that the control state remained sufficiently distanced from the hurricane treatment area.

16 The latter restriction acknowledges that hurricanes typically impact multiple nearby states. Finding null effects in our "placebo" test can also be interpreted as a confirmation of our approach's ability to control for national trends and seasonality at the state level.

4. Recessions sample

Based on the overall stability of our estimates, we next explore how our model fared out-of-sample in forecasting the unprecedented surge in initial UI claims during the Pandemic Recession that accompanied the Covid-19 outbreak, as well as during the earlier Great Recession. Combined, our results across both recessions suggest that our hurricane-based identification strategy provides a compelling alternative to previously used methods for forecasting initial UI claims.

4.1. The Pandemic Recession

As states implemented shelter-in-place policies and many non-essential businesses shut down in March 2020, unemployment insurance offices were inundated with applications. The resulting increase in new UI claims was expected to be large but the magnitude from such an unprecedented shock was unclear. Indeed, private sector forecasts ranged from 1 to 4 million claims for the week ending March 21, 2020, the first full week with restraints in place. We show that the elasticity estimate of our

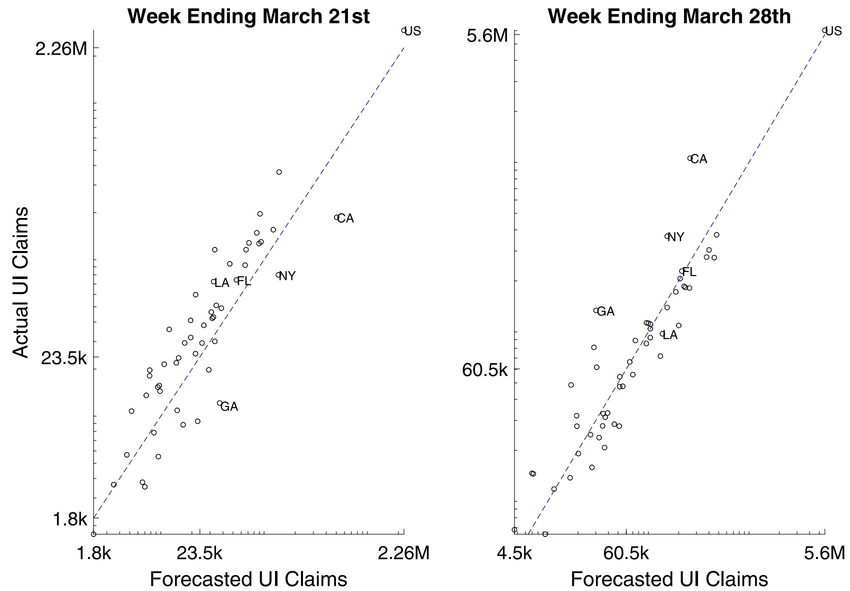

Fig. 9. Forecasted versus realized state-level initial UI claims. *Notes*: This figure reports state forecasts of UI claims against the realized values for the weeks ending March 21, 2020 (left panel) and March 28, 2020 (right panel). The solid line is the 45 degree line.

hurricane event study provided the means to translate the Google Trends unemployment topic index into an accurate forecast of the record surge, as well as the subsequent gradual decline in new UI claims into the fall of 2020.

Given the attention on the initial weeks of the pandemic, we first present the model's state (plus DC) forecasts for the weeks ending March 21 (left panel) and March 28 (right panel) in Fig. 9. While considerable variation exists across states, the model consistently produced claims forecasts that represent the largest single and two-week increase in recorded history.[17] Using the elasticity estimate from our baseline regression, the forecast for U.S. initial UI claims was 2.3 million (panel A), with a 95% prediction interval of 1.0 to 4.4 million, for the week ending March 21, 2020. The advance (and ultimately, final) release of initial UI claims was 2.9 million for the week ending March 21, essentially spot on our projection. Our national forecast for the week of March 28, 2020, was 5.7 million, compared to the final release of 6.0 million.[18]

Fig. 10 presents the full set of one-week ahead national forecasts (or, ''nowcasts'') through October 24, 2020. We also report the actual data (black line) and the 95% prediction interval implied by aggregating our state-level forecasts (shaded blue area). Our model was not only effective at capturing the unprecedented surge in initial UI claims at the beginning of the pandemic, but also the gradual decline during the summer and fall. Our largest forecast error came during the week ending April 4th, not coincidentally the week following the implementation of the CARES Act, which radically changed many aspects of the unemployment insurance programs.[19] More generally, across all of the states (plus DC) and through the first 32 weeks of the pandemic, the 95% prediction interval for the state-level nowcasts contain 96% of the actual state claims data. Moreover, none of the aggregate U.S. data fall outside of the 95% confidence interval of our aggregated state forecasts.

To benchmark the performance of our model during the Pandemic Recession, we compare its forecast accuracy against the time series model of Choi and Varian (2012), a naive random walk, and a univariate autoregressive model with one lag (i.e., AR(1)).[20] For both the Choi and Varian (2012) and the AR(1) models, the estimation sample uses weekly data from October 31, 2015, to February 8, 2020. We then project out-of-sample the 36 weeks between February 15 and October 24, 2020, re-estimating recursively through the evaluation period.[21]

17 In both weeks, more than 74% of states fall within their 95% confidence prediction intervals, with several of the states falling outside in both instances, but in opposite directions (e.g., California)–an indication that there may have been a delay in the resulting claims from search. For further details, see online appendix Table A.5.

18 By comparison, a version of the (Choi & Varian, 2012) model badly predicted both weeks. Combining early state UI reports in conjunction with a model informed by Google Trends led (Goldsmith-Pinkham & Sojourner, 2020) to predict initial UI claims of 4.2 and 5.5 million for the same weeks, an 800 thousand overestimate for the two weeks combined. See below for more details.

19 The role of the Pandemic Unemployment Assistance (PUA) program on the model's forecasting performance is investigated in Section 5.

20 On September 3, 2020, the Department of Labor revised its seasonal adjustment procedure on the national UI claims series moving from a multiplicative adjustment procedure to an additive one. To avoid having this change in the reporting standards of the forecasting variable confound our forecast evaluation, our implementation of the (Choi & Varian, 2012) model is based on non-seasonally adjusted data. Aside from this, our implementation of Choi and Varian (2012) makes the same modeling choices, including recursively re-estimating the model each week, having both the Google Trends unemployment and jobs topic series enter in levels, and conditioning on a single lag of unemployment insurance claims.

21 On June 8, 2020, the National Bureau of Economic Research (NBER) defined the peak of the business cycle to have occurred in February of 2020 and has yet to determine the trough of the Pandemic Recession.

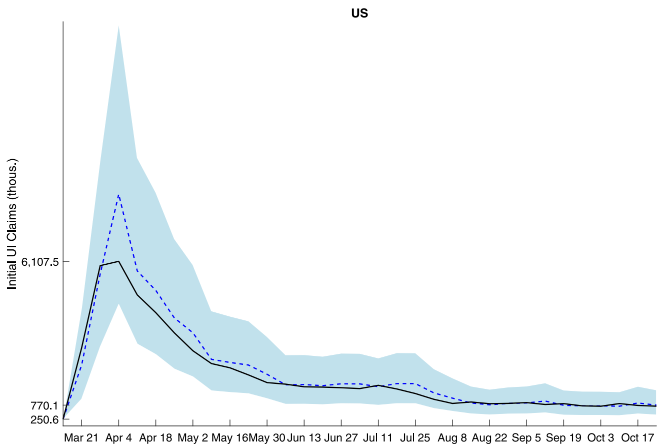

Fig. 10. Total U.S. UI claims and forecasts, week of March 14 to October 24, 2020. *Notes*: This figure reports the path of UI claim forecasts from the Google Trends model (dashed blue line) for the week ending March 14, 2020, through October 24, 2020. The shaded blue area represents the 95% prediction intervals. These intervals are generated from 10,000 bootstrap draws of the hurricane event sample and are aggregated up from individual state-level forecasts. The black line shows the final estimate of weekly U.S. initial UI claims. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3 reports the mean absolute error (MAE) of the claims forecasts from the various alternative models in column (1) and the difference in MAE relative to our model in column (2). Positive values in the second column indicate that our model's forecast is more accurate. The statistical significance of the difference in the models' accuracy is reported in column (3).[22]

During the Covid-19 pandemic (panel A), our model performs significantly better, with an MAE more than 40 percent lower, than the alternatives. Moreover, the improvements are particularly stark at the beginning of the pandemic, when there is the most uncertainty and forecast accuracy is likely most important to policymakers. Over the entire duration of the Pandemic Recession, the forecast performance gains relative to the Choi and Varian (2012) model are statistically significant at the 90 percent confidence level. Relative to the random walk and AR(1) alternatives, the forecast performance gains are statistically significant at the 80 percent confidence level. Moreover, those significance estimates are conservative. If instead, the bandwidth setting for the Bartlett kernel is set to the sample size, that is the setting proposed by Kiefer and Vogelsang (2005), the forecast gains relative to all of the alternatives are statistically significant at the 90 percent confidence level. Most importantly, given the unprecedented number of new claims during the pandemic, the size of the forecast gains are economically significant in that they represent millions of incorrectly forecasted unemployment insurance claims over an incredibly condensed period of time relative to past recessions.

Comparing the forecasts between the random walk, AR(1), and the Choi and Varian (2012) model is also instructive on the source of our model's forecast performance gains during the Pandemic Recession. During the Pandemic Recession, both the random walk model and the AR(1) model, neither of which incorporates information from Google Trends, outperformed the Choi and Varian (2012) model.[23] Moreover, during the Pandemic Recession, the forecasts from the random walk model outperformed the AR(1) by a considerable amount. This indicates that both the AR(1) and Choi and Varian (2012) model, which incorporates a lag dependent variable, could suffer from finite sample bias–an issue our hurricane event-study approach avoids.

4.2. The Great Recession

In panel B of Table 3, we show the same exercise for the 113 weeks between January 5, 2008, and February 27, 2010, that cover the Great Recession.[24] Again, our model bests each of the alternatives, although by less than the Pandemic Recession. The smaller forecast performance gain during the Great Recession is not surprising given the dramatic differences in the week-to-week fluctuations in initial unemployment insurance claims during these two recessions. Generally, we find a little more than a 10% reduction in MAE, which is still economically significant in terms of the implied number of new UI claims not predicted by these alternative models and is statistically significant at conventional confidence levels.

22 We assess statistical significance with a Diebold and Mariano (1995) test of equal mean-absolute forecast error (MAE). The p-values reported in the table use Student's t critical values and further incorporate a small-sample size correction recommended by Harvey et al. (1997). Heteroskedasticity and autocorrelation consistent (HAC) standard errors are constructed using the Bartlett kernel with bandwidth set equal to 50% of the sample size as a compromise between the standard bandwidth settings of either the number of lags used in the model plus the forecast horizon or sample size (Kiefer and Vogelsang (2005)).

23 In their forecast evaluation during the Covid-19 pandemic Larson and Sinclair (2021) include a national uni-variate AR(1) time series model as well as an autoregressive model that includes Google Trends, and find similar results in that the model with Google Trends does worse out-of-sample than the univariate AR(1) model.

24 More specifically, we estimate our version of Choi and Varian (2012) using March 5, 2005, to December 29, 2007, and recursively out-of-sample forecast the Great Recession.

Table 3
Forecasting performance comparison: Pandemic recession & great recession.

	MAE	Difference w/ ABBFSS (2020)	DM p-value
	Panel A: Pandemic recession (36 weeks)		
Random walk	0.180	0.085	0.196
Choi and Varian (2012)	0.444	0.349	0.097
AR(1)	0.315	0.220	0.153
	Panel B: Great recession (113 weeks)		
Random walk	0.082	0.009	0.025
Choi and Varian (2012)	0.099	0.026	0.002
AR(1)	0.088	0.015	0.004

Notes: This table reports the mean absolute forecast error (MAE) for both the forecasting model using the elasticity estimate from the sample of hurricanes, the traditional time series model of Choi and Varian (2012), the AR(1) model, and the naive random walk forecast for both the Covid-19 Pandemic Recession (panel A) and Great Recession (panel B). We assess statistical significance with a Diebold and Mariano (1995) two-sided test of equal mean-absolute forecast error. The p-values reported in the table use Student's t critical values and further incorporate a small-sample size correction recommended by Harvey et al. (1997). Heteroskedasticity and autocorrelation consistent (HAC) standard errors are constructed using the Bartlett kernel with bandwidth set equal to 50% of the sample size as a compromise between the standard bandwidth settings of either the number of lags used in the model plus the forecast horizon or sample size as suggested by Kiefer and Vogelsang (2005).

Here again, the AR(1) and random walk, which do not incorporate Google Trends data, outperform Choi and Varian (2012), an observation in Choi and Varian (2012) as well. And again, the random walk outperforms the AR(1) model. Taking the results across both time periods, it is clear that the naive random walk model is the preferred forecasting approach amongst the set of alternatives that we consider beyond our model. But our hurricane event study approach can incorporate the informational content embedded in Google Trends to notably improve on the naive random walk. This performance likely reflects leveraging the hurricane event-study research design to overcome the inherent difficulties associated with state-level UI claims data.

5. Google Trends and the Pandemic Unemployment Assistance (PUA) program

The analysis thus far has indicated that our event study research design provides a credible method to forecast initial UI claims with Google Trends during recessions. A critical aspect of this design was that it allowed estimating an elasticity that was unlikely to confound "news"-related search activity of future labor market outcomes with search related to the real-time demand for filing a UI claim. The Covid-19 pandemic, given its global scope and magnitude, generated a significant amount of labor market-related news. While our identification strategy for regular state program UI claims produces forecasts that appear to be mostly robust to these concerns, this may not be the case in all instances related to unemployment insurance. Most notably, it remains an open question for the Pandemic Unemployment Assistance (PUA) program, which was passed as part of the CARES Act on March 28, 2020, and greatly increased the set of individuals eligible to file for unemployment insurance for the first time, primarily by adding independent contractors and other "gig" economy workers.[25]

This new UI program potentially affects our model's forecasts in at least two ways. Perhaps most importantly, the introduction of the PUA program led to a large increase in the overall number of UI claims. This is important for interpretation since our hurricane model is estimated in a setting without a PUA program. By contrast, during the pandemic, someone searching for information to file a claim for unemployment insurance in a regular state UI program is likely to engage in search behaviors that are similar to another person who ultimately will file a claim under the PUA program. Those distinctions can lead the model to overestimate initial UI claims related to the regular state UI programs. These issues are further complicated by the lack of standardization in the way states report PUA claims: some report regular and PUA claims together and others separate them. Moreover, the very novelty of the PUA program could further contribute to a deterioration in our model's forecasting performance. Not only were a much larger set of individuals newly eligible for UI during the pandemic, but many others, including policy experts, reporters, researchers, industry experts, and interested members of the public, were likely engaging in unemployment topic-related search on Google for information on the new program that was not likely to lead to an eventual unemployment insurance claim.

Fig. 11 presents evidence that both of these effects could have affected our model's forecasting performance. We compare the forecasted cumulative log growth between March 14 and October 24, 2020, of state-level claims against either the cumulative growth of initial UI claims coming only from regular state UI programs (left panel) or the cumulative growth of initial UI claims coming from both regular state UI and PUA programs (right panel).[26] The 45-degree line is marked in dashed blue. A comparison of the left and right panels makes clear that our model's forecasting performance is influenced by

[25] In addition to the PUA program, there was also the Pandemic Emergency Unemployment Compensation (PEUC) program. This program has led to many fewer claims than the PUA program and for data availability reasons is beyond the scope of our analysis.

[26] Importantly, the cumulative forecasted growth rates represent the full path of forecasts conditional on each state's path of Google Trends during the pandemic, as opposed to the "nowcasts" reported earlier which incorporated the prior week's information on new UI claims.

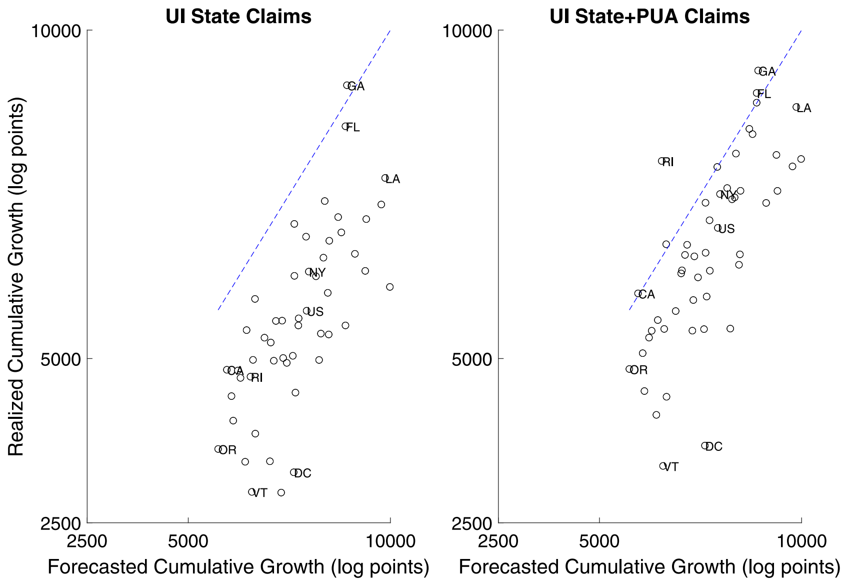

Fig. 11. Forecast and realized cumulative growth, March 14 to October 24, 2020. *Notes*: This figure reports the forecast of the cumulative growth in initial UI claims between the week of March 14th and October 24, 2020, from the Google Trends model versus the realized cumulative growth over those same weeks across all states and D.C. In the left panel, only regular state program UI claims are used for realized growth values, while in the right panel the initial claims that came from both regular state programs and a state's pandemic unemployment assistance (PUA) program are used for the realized growth values.

Table 4
Elasticity estimates with and without PUA claims, Pandemic Period.

	Dependent variable:			
	State claims		State + PUA claims	
	(1)	(2)	(3)	(4)
Google trends	1.288	0.822	1.211	0.899
	0.075	0.097	0.069	0.092
Week FEs		X		X
Adj. R^2	0.518	0.755	0.521	0.657
N	1632	1632	1632	1632

Notes: This table reports estimates of the elasticity of UI take up with respect to Google Trends search using a panel of weekly state (and DC) claims and Google Trends data but estimated only over the 32 weeks of the pandemic sample. In the first two columns, the regular state program initial claims are used. In the last two columns, the PUA claims are added on. Week-of-year fixed effects are included in columns (2) and (4). Standard errors are clustered at the state level.

whether or not PUA claims are included. Generally speaking, the forecasts are better when the new PUA claims are included. However, across both panels, the propensity of the cumulative growth forecasts to overestimate what has, thus far, been realized as new UI claims is suggestive that some "news-related" search activity still may have made its way into our estimates, particularly during the initial weeks of the pandemic when our forecasts were typically too large.[27]

To provide more evidence of these news effects, we re-estimate our UI elasticity over the Covid-19 period, rather than around hurricanes, using a state-level panel that excludes (columns 1 and 2) and includes (columns 3 and 4) PUA claims. To identify the scope for "news"-related search, we estimate the regressions with (columns 2 and 4) and without (columns 1 and 3) week-of-year fixed effects. With week-of-year fixed effects, the variation used to identify the elasticity comes from the differential timing of when states experience surges in demand for unemployment insurance. By contrast, the national surge in unemployment topic-related search that was likely partially driven by the introduction of the PUA program early on in the pandemic would be included in the statistical models that do not include week-of-year fixed effects.

Table 4 reports the results. Without PUA claims (columns 1 and 2), we find an estimated elasticity of 1.29 (standard error of 0.08) without week-of-year fixed effects and 0.82 (0.10) with them. Including PUA claims (columns 3 and 4), leads to an estimated elasticity of 1.21 (0.07) without week-of-year fixed effects and 0.90 (0.09) with them. Relative to the hurricane-based elasticity of 1.06, the the specifications that include the PUA claims appear to do better and indeed the specification that accounts for week fixed effects is the only one that fails to reject our baseline estimate. Together, these results indicate that our model was flexible enough to capture the unemployment-related search activity that resulted in PUA claims; and while some of this search-related activity was certainly driven by a "news" component, most likely from the PUA program itself, the approach still provided an elasticity estimate that was otherwise likely to lead to accurate forecasts.

Our model can also help analyze other aspects of the UI system. For instance, it could be used to alert policymakers of the impact of changes in the state's UI processing

27 In late April 2020, "unemployment" was the most searched term on Google in almost every county of the country. See https://twitter.com/i/status/1268604487482011649.

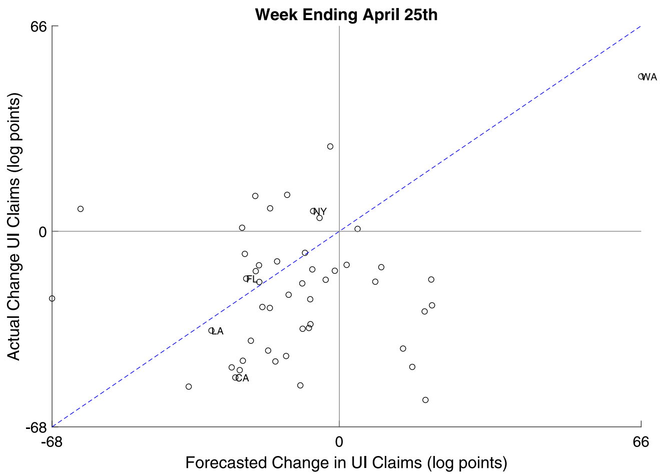

Fig. 12. Cross-section of state-level UI claims forecasts and realizations, week of April 25, 2020. *Notes*: This figure reports the cross-section of state-level forecasts and realizations of UI claims for the week ending April 25, 2020. The dashed blue line is the 45 degree line.

infrastructure, not unlike the episode that occurred during the week ending April 25, 2020. That week, our model predicted that the state of Washington would depart substantially from the national experience and have a 66 log point increase in claims (see Fig. 12). This forecast was mostly realized; there was a surge in new UI claims in Washington that came about as a result of the drastic overhaul of the unemployment insurance department's website. Google Trends picked this event up; more traditional forecasting models would have had difficulty detecting it in real time.[28] This is the kind of event that need not occur only during a recession, suggesting that our research design could have broader uses such as projecting staffing needs during important changes to rules and processes.

Additionally, even the "misses" of our model might be of interest, as they might indicate problems in the UI system, such as backlogs, processing issues, or fraud. For example, California began *withholding* its weekly UI claim reports beginning the week ending September 26, 2020, on the heels of a surge in potentially fraudulent PUA claims in late August (see red crosses in Fig. 13 which plots growth rates relative to March 14, 2020).[29] Not coincidentally, those extra PUA claims were not supported by our model's California projection (e.g., see dashed blue line in Fig. 13 which plots growth rates of Google Trends unemployment topic search activity in California relative to March 14, 2020). While the full scope of the number of fraudulent claims still is unclear, the experience in California suggests that even the forecast *errors* of our model could be useful to policymakers.

6. Discussion

By producing a forecast from state-level UI claims data based on a well-identified elasticity between Google Trends search activity and unemployment insurance take-up, our hurricane event-study design was flexible enough to accommodate both the unprecedented surge in UI claims and eligibility changes that were made during the Covid-19 pandemic in ways that previous forecasting models that used Google Trends could not. The fact that this was the case at both the state and national levels, and that the model performed well during the Great Recession as well, should only increase its value in a host of other potential applications, including the growing need to develop procedures to accommodate the unprecedented recent changes that have occurred within state UI systems.

Furthermore, as one of the most timely measures of labor market activity, UI claims are regularly incorporated into economic activity indexes and forecasting models (e.g., Lewis et al. (2020), Diebold (2020), Brave et al. (2020)). Our results indicate that perhaps these models should also consider including Google Trends as a complementary source of high-frequency information on labor market activity. We also encourage future research to investigate the extent to which the research design implemented here can help inform and improve upon the forecasting ability of these models, particularly in settings where there is substantial value in capturing abrupt turning points. The use of carefully identified relationships between state-level data and Google Trends could potentially have broader implications beyond UI claims, including employment, unemployment, personal income, and GDP.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

[28] See https://www.spokesman.com/stories/2020/apr/20/clunky-state-website-struggles-with-record-applica/.

[29] For the full report explaining the temporary halt in reporting unemployment insurance claims, see https://www.govops.ca.gov/wp-content/uploads/sites/11/2020/09/Assessment.pdf.

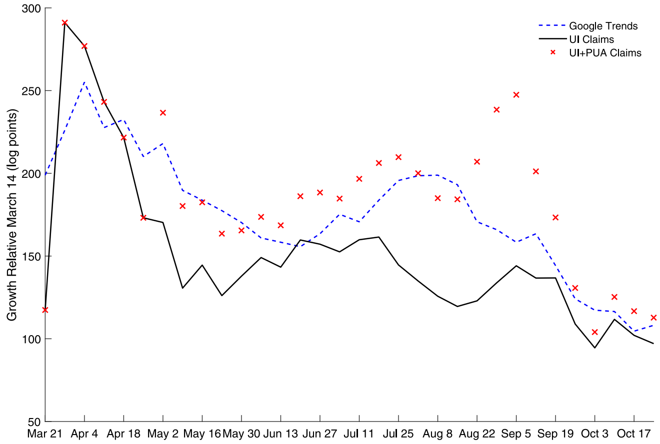

Fig. 13. UI claims with and without PUA and Google search, California. *Notes*: This figure reports California's relative log growth of initial UI claims in the regular state program (solid black), regular state claims and Pandemic Unemployment Assistance (PUA) claims (red crosses), and Google Trends search intensity for the unemployment topic (dashed blue) for the weeks from March 21st, 2020 to October 24th, 2020. Growth rates are relative to the week of March 14, 2020. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2021.04.001.

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